THE CAT AT CHRISTMAS VIDEO

The conditional distributions used for the sampling experiment were as shown below.

		FROM	
		Outside	Inside
TO	Outside	0.8	0.4
	Inside	0.2	0.6

In the sampling trial, there were 110 samples, and of these 10 were declared to be 'burn-in' – my choice, there are tools to help but this is a choice of the experimenter. All the sample values are shown below, with 1 for 'inside' and '0' for outside.

	Burn In	0	10	20	30	40	50	60	70	80	90
1	1	0	1	1	0	0	0	0	0	1	0
2	1	0	1	0	0	1	0	0	0	1	1
3	0	0	0	1	0	1	0	0	0	0	0
4	0	1	0	0	0	1	1	1	0	0	0
5	1	1	0	1	0	1	0	0	0	0	1
6	1	0	0	1	1	1	1	0	0	1	1
7	0	0	0	1	1	0	0	0	0	1	0
8	0	0	0	0	1	0	0	0	0	0	0
9	0	1	1	0	0	0	0	1	1	0	0
10	0	0	1	0	0	0	1	0	0	0	0

Ignoring the 'burn-in', 33/100 were 'inside'. Therefore, the MCMC estimate of the probability the cat is inside on Christmas Day, i.e. a long time after the starting position, is 0.330. I can do fancy maths on this nice simple problem and the answer is 1/3 i.e. 0.333. Very close! I could do a longer chain and I would expect the answer to get more and more precise: where we stopped was my choice over how much time to spend on generating it. I could monitor the value after a long period and see how sensitive my estimate is to a longer chain.

One of the advantages of MCMC is the richness of the information that can be obtained. For example, "what is the probability of the cat being inside >3 times in a week?" To answer this I could start at sample 0+7 which is 0, and look at the previous 7 days (inclusive of day 7) and see if my sample chain had the cat in >3 days. So days 1-7, the answer is 'No' – the cat was only inside for 2 days; days 2-8, 'No'; … ; days 19-25, 'Yes'; days 20-25, 'Yes'; … ; days 94-100, 'No'. I have 14/94 'Yes', so the probability is 0.149.

Consecutive samples in my 94 values tend to be correlated – e.g. the 14 'Yes' come in 1 run of 5 (days 25 to 29), one run of 8 (days 52 to 59), and a single event ending on day 87, which seems too unlikely an occurrence for independent sampling. I could change my assessment and look at days 1-7, days 8-14 etc. This is 'thinning' to every 7th sample as I am throwing away some intermediate values because I don't wish to deal with the correlations, however I now only have 14 realisations, and of these 3 are 'Yes and I have the result 0.214. However this is now estimated with only a sample of 14, so to get large samples with less correlation I need a longer chain.

Running a chain of 99900, after a burn-in of 100: the full chain gives an estimate of 0.3326 for P(Cat is Inside), and 0.2535 for P(Cat is Inside >3 times in a week). If thinned by every 7th sample the probabilities are 0.3342 and 0.2533 respectively, with approximate s.e.s are ~ 0.0039 and 0.0036 respectively. Assuming (wrongly) that the binomial can be used for the estimate for P(Cat is Inside >3 times in a week) = 0.1733, notably less than the true value.

Note: no inference is intended on the differences between the thinned and un-thinned means; however, thinning is helpful in establishing the full distributional properties of the chain and the confidence in any inference from it, however this is beyond the scope of QGM.